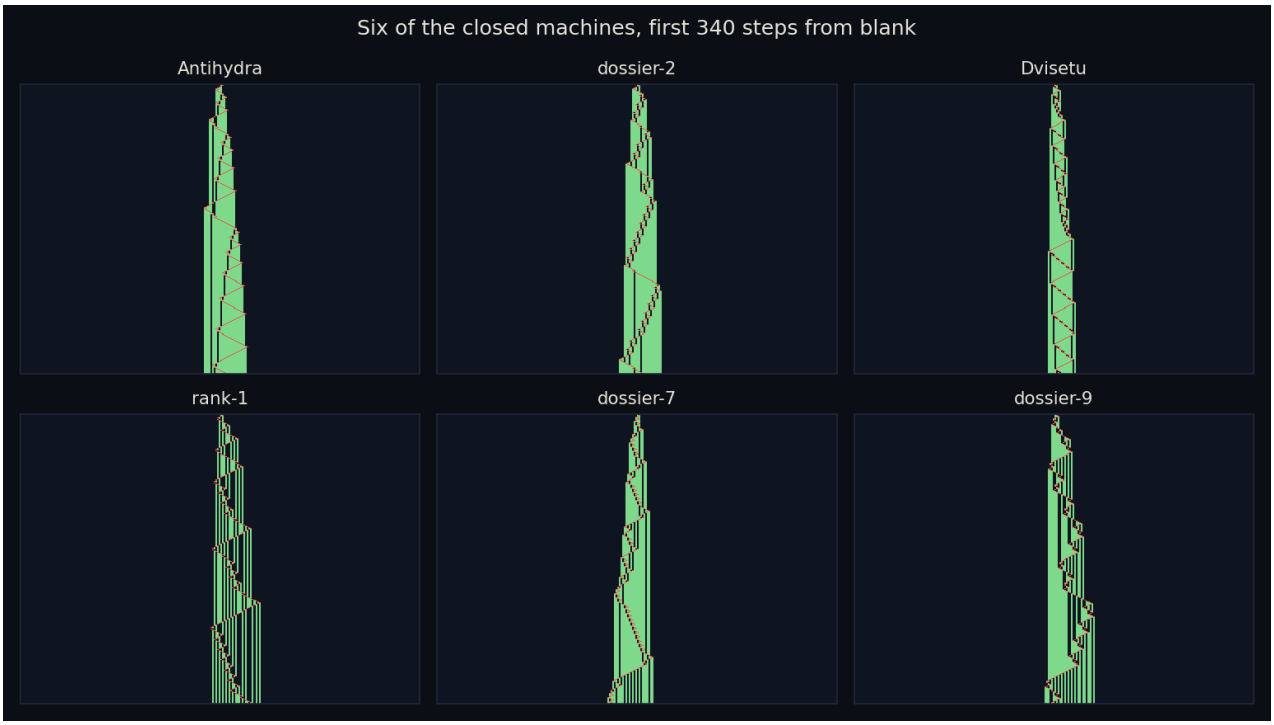
The Closed Machines of BB(6)

Nine kernel-checked conditional closures, machine by machine: what each cryptid is, how it was closed, and what remains open

Enan Srivastava (with an AI assistant; every proof checked by the Lean 4 kernel)

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The busy beaver problem at six states is blocked by a residual set of 1,094 machines that defeat every known automated decider; the community calls them cryptids, because each hides an iterated arithmetic map of the Collatz kind. This document is a field guide to the nine that have been *closed*: for each, a machine-checked theorem states that if one explicit arithmetic conjecture holds, the machine never halts from the blank tape. Every theorem is verified by the Lean 4 kernel at the minimal axiom set $\{\text{propext}, \text{Quot.sound}\}$ with no `sorry`. All nine of the program's exact-law machines are now closed.



The method, common to all nine



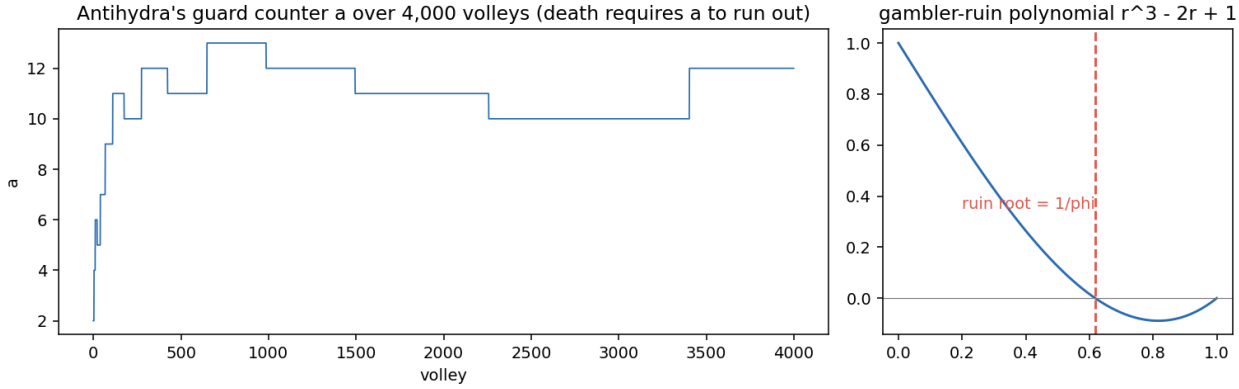
Watch. Each machine has a natural clock: a recurring configuration (a record, a spawn entry, an arrival) at which the tape has a bounded description. Five integers suffice for dossier-2/3; a triple for Antihydra; one hole and two counters for Dvisetu; a word of blocks for the rank-1 machine. **Compress.** The exact law of that description is derived empirically and fail-closed: every transition and its exact step count must be reproduced, and any miss aborts the pipeline. Laws are frozen, then validated on held-out horizons. **Certify.** The law is compiled into Lean: raw semantics; a *corridor lemma* showing every halt funnels through a single probe pattern; episode theorems with exact step counts; inductions over the unbounded middles; a concrete startup anchor; and a transport theorem showing the abstract orbit predicts the machine forever. The result, per machine: $\mathsf{Safe} \to \neg\mathsf{RawHalts}$, and (where tested) the sharpness converse: states violating the conjecture provably halt within tens of steps.

The engineering matured across the closures. The early files (dossier-2/3, 6,000–9,000 lines) prove gait-level episodes; the newest (dossiers 7/8, about 1,000 lines each) run their certified engines at single-raw-step grain, where every branch of the record law, including residue cases never observed at any empirical horizon, is simulated exactly with no assumption folded anywhere.

Antihydra

code	`1RB1RA_0LC1LE_1LD1LC_1LA0LB_1LF1RE_---0RA`
status	closed (conditional); the community's best-known cryptid
state space	hydra triple (a, m, c); tape $1^a\,0\,1^m\,0\,1^c$
keystone	`safeOrbitA_nonhalting` (2,130 lines)
conjecture	`SafeOrbitA`: the hydra orbit never dies
halt corridor	F entered only via $E0$, moving left
validation	mirror exact vs. 51.8M raw steps (10,001 records)

Antihydra iterates a hailstone-like map in which a guard counter a drifts upward on average while occasional merges spend it. Death requires the counter to run out, and the walk is a gambler's ruin with steps $-1/+2$ whose ruin probability is the root of $r^3 - 2r + 1$: exactly $1/\varphi$. The golden ratio appears in Antihydra's *hazard* where dossier-3 carries it in its *cone*: same constant, different organs. Two further facts with closed forms: the realizable dying family $(1, 2, c)$ halts in exactly $(3c^2 + 44c + 125)/4$ raw steps for odd c, and the abstract death cone grows *linearly* (branching constant exactly 1, backward merges throttled by the $a \geq 2$ guard), the third data point in the program's branching-constant series $\{\varphi,\ 2\text{-adic},\ 1\}$.



Dossier-1

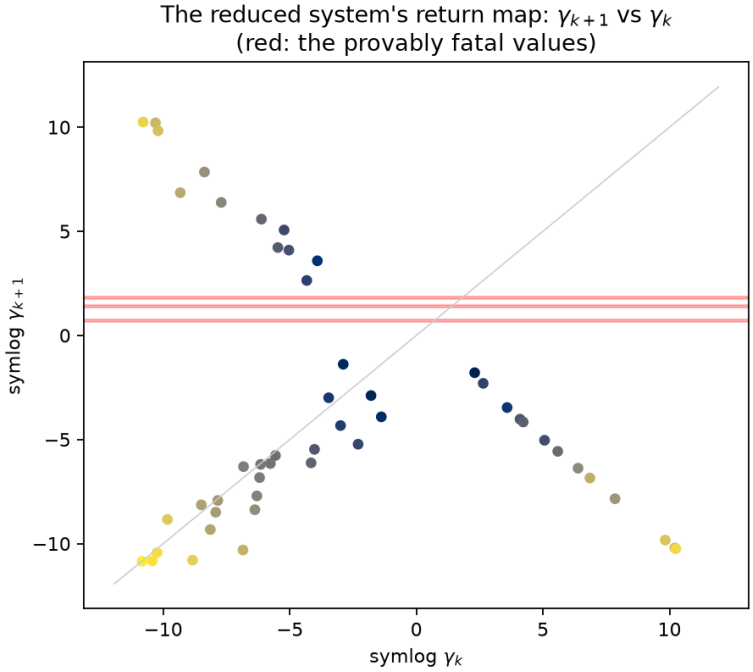
code	`1RB0LE_1LB1RC_0RF0RD_1RE0LA_1LA1RE_---0RB`
status	closed (conditional)
state space	a stack of levels (chain orbit)
keystone	`dossier1_nonhalting` (5,770 lines)
conjecture	the abstract chain orbit never dies
halt corridor	`halt_requires_c_probe_00`

The program's first original closure and the template for everything after it: the layered endgame (halt localization, then an invariant discharged by a raw-to-macro closure induction) that later files industrialized. Its 33 macro obligations were first discharged empirically at 10^8 steps, then certified.

Dossier-2 and dossier-3 (the mirror pair)

codes	`1RB1RC_1LC1LC_1RE0LD_1LB1RF_---1RA_0RD1RD` `1RB1RB_1LC0RE_---1LD_0RA1LB_1RA1LF_0LE1LE`
status	closed (conditional); one law, two machines
state space	five integers: block ends l, r, holes h_1, h_2, tag
keystones	`dossier2_nonhalting` (6,130 lines), `dossier3_nonhalting` (9,020)
conjecture	`SafeMetaOrbit`: the five-integer iteration never enters an explicit fatal set
halt corridor	the probe event: C reads 0 with a 0 to its right
validation	predictor exact over 10^9 raw steps per machine; seven episode theorems with exact step counts (crawl 10, cross 13, terminate 13, fuse 19, …)

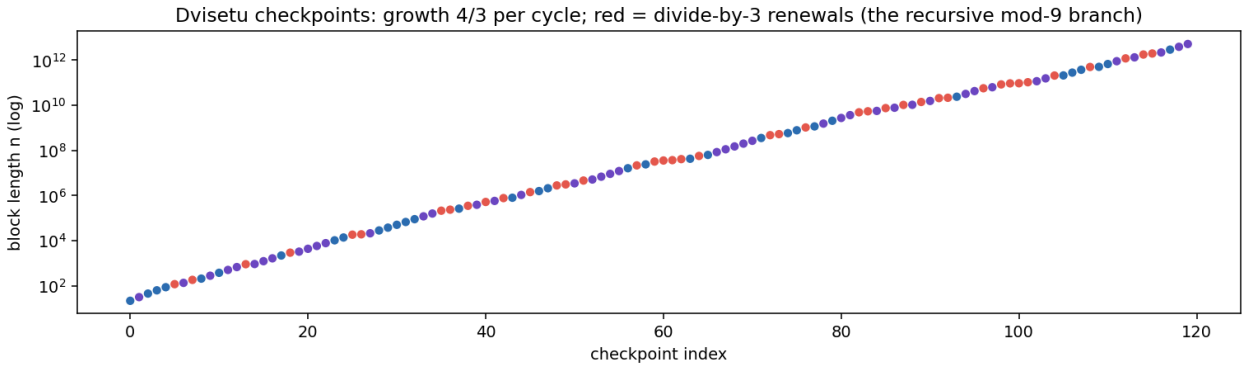
The flagship reduction. At every spawn entry the tape is a solid block of 1s with exactly two holes; seven certified episodes cover everything the machine does. Sampling at renewals collapses the dynamics to a closed-form two-counter map whose fatal set is a width-five window around the diagonal; the orbit avoids it for 10^8 iterations (about 10^{15} raw steps) with minimum margin 4. The proof routes that fail are themselves mapped: no congruence invariant to modulus 384, factor complexity $\sim 1.6^n$ (not a rotation), and 33,551 ranking-class halfplane families exhaustively refuted on the true fatal model. Dossier-3's abstract death cone is an exact Fibonacci tree ($L(k) = L(k-1) + L(k-2)$, branching at φ, mechanism: no terminate follows terminate).



Dvisetu (dossier-6)

code	`1RB0RB_1LB0LC_1RD1LC_1RE0RA_0RF1RA_---1LB`
status	closed (conditional); this program's own miner candidate (July 2026), re-surfaced blind by the deep sweep
state space	one hole, two counters: $1^a\,0\,1^b$, plus checkpoint scalars 1^n
keystone	`safeDvisetuOrbit_nonhalting` (1,942 lines)
conjecture	`SafeCheckpointOrbit` (orbit guards + the divide episode, folded as `DivideFaithful`)
halt corridor	E reading 0 with a 0 to its *right*
validation	2,746/2,746 timed record transitions at 2×10^7 steps; divide sub-rule 5/5 to $n = 18{,}648$

The simplest state geometry in the program, and a genuine generalized-Collatz specimen: drift $(a, b) \to (a{+}4, b{-}3)$ at exactly $(32b - 41)/3$ steps per record, collapse to a single block, then a renewal branched on n mod 3, whose $n \equiv 0$ branch *divides by* 3 recursively (divide while the quotient is 0 mod 9, base case by the residue mod 9). The drift gait resolved into four passes over a five-step motif, making the step count integral by construction. From dossier opening to kernel-checked closure: about 36 hours.

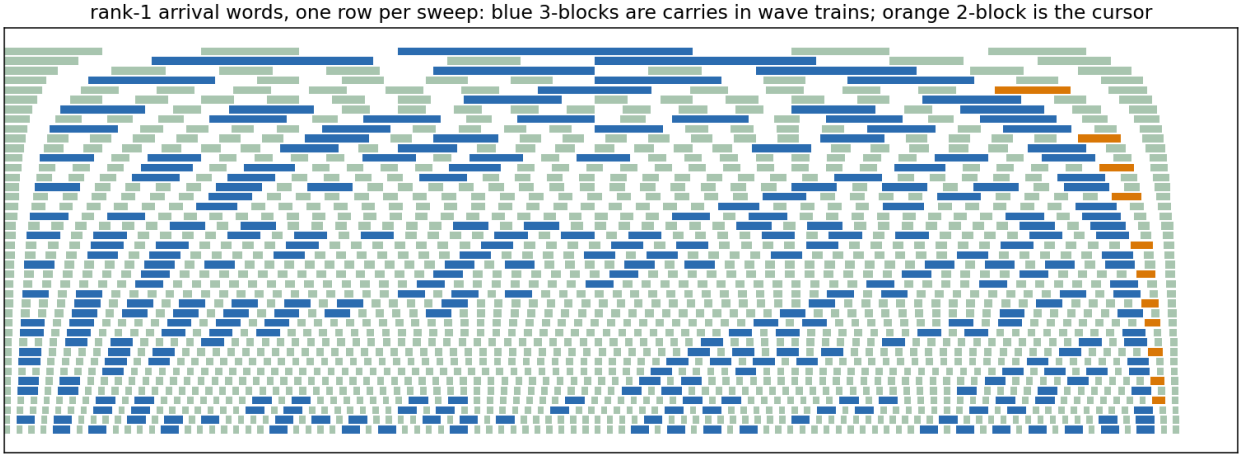


Rank-1 (dossier-5): the first growing-dimension closure

code	`1RB1RD_1LC0LC_0RD0LB_1RF0LE_0RD---_0RA1RC`
status	closed (conditional); first closure on a state space of growing dimension
state space	a *word* of 1-blocks (lengths 1/2/3, single-0 gaps) that grows without bound
keystone	`safeWordOrbit_nonhalting` (2,126 lines), zero folded assumptions
conjecture	`SafeWordOrbit`: the word orbit never dies (a single clause)
halt corridor	D reads 1 with a 1 to its left
validation	calculus frozen-exact 2,179/2,186 at 2×10^8 steps; reseed differentials exact; Lean mirror realizes every arrival configuration at the predicted cumulative time over 2.2M steps

Semantically a two-phase odometer on $\{1, 3\}$-words: the 3-blocks are carries traveling in wave trains, a deposited 2-block is a cursor that restarts the next pass at its own position, and the boundary rule is a literal block trisection $3 \mapsto 1, 1, 1$. Its death tree branches 2-adically in train lengths (the number of one-sweep preimages of a word is exactly $\prod_i 2^{\max(0,t_i-2)}$), where dossier-3's branches at φ. The certified layer works at cell grain, one level below the learned calculus, where the boundary families and the decision function are subsumed and proven.

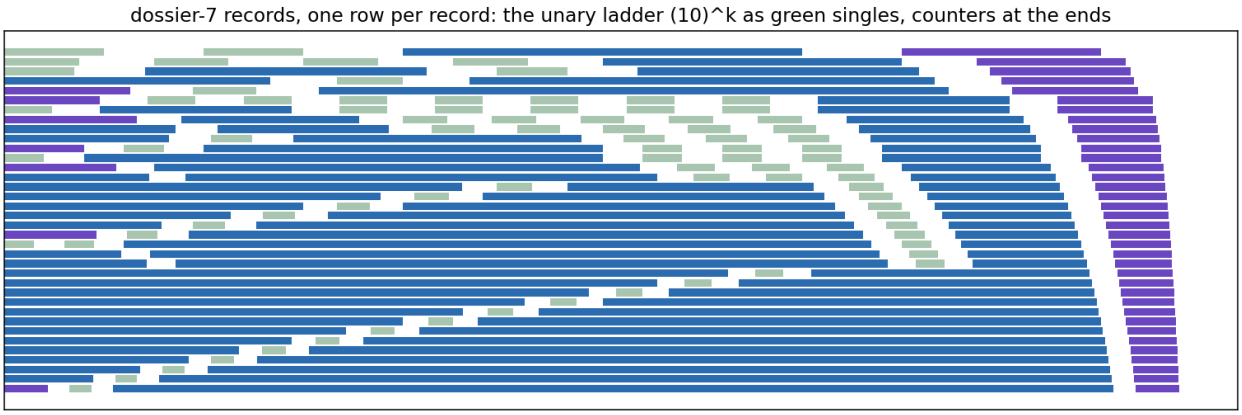
rank-1 arrival words, one row per sweep: blue 3-blocks are carries in wave trains; orange 2-block is the cursor



Dossiers 7 and 8: the raw-step-grain pair

codes	`1RB1LA_1LC0RE_1LF1LD_1RE0LC_1LB1RA_---0LA` `1RB0LC_1LC0RD_1LF1LA_1LB1RE_1RB1LE_---0LE`
status	closed (conditional); the strongest engineering of the set
state space	unary-ladder words $1^a\,0\,1\,0\,1^b\,0\,(1\,0)^k\cdots$ growing left; counters (a,b,k) and (a,b,k,g)
keystones	`safeD7Orbit_nonhalting` (1,036 lines), `safeD8Orbit_nonhalting` (1,037)
conjecture	the word orbit never dies (single clause each)
halt corridor	F entered only from $C0$; probe: C reads 0 needing a 1-guard to its left
validation	design mirrors 5,263/5,263 and 3,890/3,890 record transitions at 2×10^7 steps, validated before any Lean was written; 20,000 randomized episode plants each

Sibling machines with opposite ladder parity (and, for dossier-8, an epoch counter). Their certified engines run at single-raw-step grain: every raw move is one of three proven episodes on a boundary pair, so *every* branch of the 13- and 12-rule record laws, including residue cases whose first occurrence is projected near 3×10^{12} steps, is simulated exactly and proven faithful. Nothing is folded; the only fail-closed exits are the genuine halt patterns and fuel.



Dossier-9

code	`1RB0LF_1RC---_1LD0RF_1RA0LE_0LD1LD_0RD0RB`
status	closed (conditional); the ninth and final exact-law machine
state space	words on a right-growing frontier; parametric calculus at 100.00% held-out (271,344/271,344 segments at 8×10^6 steps) preceded the closure
keystone	`safeD9Orbit_nonhalting` (1,059 lines), single-raw-step engine, zero folds
conjecture	`SafeD9Orbit`: the word orbit never dies
halt corridor	the simplest of the set: the fatal probe *is* the halt (B reads 1; B entered only via $A0$ or $F1$)
validation	mirror 5,804/5,804 record transitions exact at 2×10^7 steps; fuel margin 2,039×; 20,000 randomized episode plants

What the conjectures have in common, and what remains

Every closure ends at a Collatz-class orbit-avoidance statement, and the program has mapped why no shortcut exists: a sound invariant search over every learned law in the cohort (congruence closures to modulus 64, monotone functionals, two independent implementations that first cross-audited each other) found *zero* invariant-decidable machines, and ranking-class certificates were exhaustively refuted on the one system with a complete fatal model. The orbits avoid death for measure-theoretic reasons (drift, uniform hazard, Borel–Cantelli tails, seed-space density $\approx 12/X$), and finite certificates cannot see measure-theoretic truths. The standing deep project is the invariant-measure route: the renewal maps here are explicit, total, and piecewise affine with verified uniform hazard, materially better conditioned than Collatz itself.

Verify everything yourself

```
git clone https://github.com/Enan456/bbchallenge-proofs
cd bbchallenge-proofs/busy-beaver-core && lake build
# the build prints, per keystone theorem:
#   depends on axioms: [propext, Quot.sound]
```

External approval path: Palomar registration (entries pinned and validated; one browser session), bbchallenge forum publication with invited reproduction (post drafted), and DeepMath 2026 review (extended abstract at v19). No process makes these unconditional; approval means externally reverified reductions, each machine's halting problem restated as one explicit arithmetic question.